Module 03 | Saturday Quick Reference

Onsite: 9:00 AM-5:00 PM | Jupyter and Colab | Synthetic student data

01 A Quick Model Workflow Recap

Meaning: A supervised model maps input features to a known target. The training portion teaches the pattern; held-out test records check whether the pattern generalizes.

Remember: Every model comparison today uses the same split. Otherwise differences may be caused by easier or harder test examples rather than the algorithm.

Functions and terms: Feature matrix X - Input columns; Target y - Known pass/fail outcomes; Test metric - Quality on unseen held-out rows

Common mistake: Do not compare different test splits and then claim the highest score came from the better model.

02 Linear Regression — Predict a Number

Meaning: Linear regression predicts a numerical target using a weighted sum of inputs. For one input, the model is a line: predicted score = intercept + coefficient × study hours.

Remember: The fitted line summarizes a relationship rather than memorizing each row. We use it as a simple, interpretable first regression model.

Functions and terms: Coefficient - Change in predicted output per unit of input; Intercept - Predicted value when input equals zero; Residual - Actual value minus predicted value

Common mistake: The line is not proof that studying causes exactly that many marks. Real outcomes depend on many factors.

03 Regression Errors: MAE, MSE and RMSE

Meaning: Regression errors measure how far predicted numbers are from true values. MAE averages absolute errors. MSE squares errors. RMSE is the square root of MSE.

Remember: MAE is easy to explain in the original units, while RMSE penalizes large misses more strongly. We choose metrics based on what kinds of errors matter.

Functions and terms: MAE - Average |actual - predicted|; MSE - Average squared error; squared units; RMSE - Square root of MSE; original units

Common mistake: Never report MSE as if its unit were marks; the units are marks squared. RMSE returns to marks.

04 R^2 and the Mean Baseline

Meaning: R^2 compares the model against a baseline that predicts the average training outcome. $R^2 = 1$ is perfect; $R^2 = 0$ matches the mean baseline; negative means worse.

Remember: A model can have a low average error yet still be weak if a simple baseline does just as well. Compare both error and R^2 on held-out rows.

Functions and terms: $R^2 = 1$ - Perfect on the evaluated data; $R^2 = 0$ - Similar to predicting a constant mean; $R^2 < 0$ - Worse than the evaluated mean baseline

Common mistake: R^2 does not prove causation or reliability outside the data range. Negative values are possible and meaningful.

05 Logistic Regression — Predict a Category

Meaning: Despite its name, logistic regression is usually a classification algorithm. It calculates a score, transforms it to a probability and applies a threshold to choose a class.

Remember: For a binary pass/fail target, it predicts the estimated probability of passing. Probabilities help us choose a different decision threshold when needed.

Functions and terms: predict() - Returns 0 or 1 labels; predict_proba() - Returns per-class probabilities; Threshold - Cutoff, often 0.50, for choosing class 1

Common mistake: Do not fit and evaluate on the exact same rows. max_iter allows more optimization steps; it does not guarantee a better model.

06 Confusion Matrix — Four Outcomes

Meaning: A confusion matrix counts correct and incorrect classifications. For class 1 = passed, it separates true positives, true negatives, false positives and false negatives.

Remember: A single accuracy number hides which students are missed. False negatives may matter more when the purpose is early support for at-risk learners.

Functions and terms: TP / TN - Correct positive / negative decisions; FP - Predicted positive, actually negative; FN - Predicted negative, actually positive

Common mistake: Always state which class is positive. The meaning of FP and FN changes if class 0 is treated as the positive event.

07 Accuracy, Precision, Recall and F1

Meaning: Accuracy is the share of correct predictions. Precision checks how many predicted positives are truly positive. Recall checks how many actual positives we find.

Remember: F1 is the harmonic mean of precision and recall. It is useful when both false positives and false negatives matter, especially with class imbalance.

Functions and terms: Accuracy - All correct / all predictions; Precision - TP / (TP + FP); Recall - TP / (TP + FN)

Common mistake: Accuracy alone is unsafe when one class is much more frequent. Always inspect target balance and a confusion matrix.

08 Class Imbalance and Thresholds

Meaning: Class imbalance means one outcome is much more common than another. Predicting only the majority class can look strong under accuracy while failing at the rare class.

Remember: A threshold trades precision against recall. Lowering the cutoff usually predicts more positives, increasing recall but possibly increasing false alarms.

Functions and terms: Majority baseline - Always predicts the most frequent label; Class weight - Give extra weight to less frequent class; Decision threshold - Probability cutoff for a positive prediction

Common mistake: Do not choose the cutoff by repeatedly checking test results. Tune on validation data and reserve test data for final evaluation.

09 Decision Trees — A Chain of Questions

Meaning: A decision tree repeatedly asks questions such as “attendance <= 70?” and sends records down branches until a leaf produces a prediction.

Remember: Shallow trees are easy to explain but may miss patterns. Deep trees can memorize training examples and perform poorly on new rows.

Functions and terms: Split - A question used to divide records; Leaf - The end node containing a prediction; max_depth - Maximum number of levels of splits

Common mistake: max_depth=None may fit much more detail than the data can justify. Compare training and test scores, not only training score.

10 Random Forest — Many Trees Vote

Meaning: A random forest trains many decision trees on different sampled rows and subsets of features. Their votes are combined for classification.

Remember: A single tree can be unstable. Averaging many trees usually reduces sensitivity to small data changes, though the result is harder to explain fully.

Functions and terms: n_estimators - How many trees are trained; max_depth - Limits how complex each tree becomes; random_state - Makes random choices reproducible

Common mistake: More trees do not guarantee improved accuracy and require more work. Start with sensible defaults and compare to the baseline.

11 Overfitting, Underfitting and Generalization

Meaning: Underfitting happens when a model is too simple to learn important patterns. Overfitting happens when it memorizes training details rather than general patterns.

Remember: Generalization means doing well on new data. Compare train and validation scores to understand whether a model is learning broadly useful patterns.

Functions and terms: Underfitting - Low train and low test performance; Overfitting - High train but lower test performance; Generalization - Good performance on unseen rows

Common mistake: A good result is not simply the largest training score. Choose a model that stays reliable on unseen data.

12 Cross-Validation and Error Analysis

Meaning: Cross-validation repeatedly splits training data into folds. Each fold gets a turn as validation while the remaining folds train the model.

Remember: Averaging fold scores provides a more stable view than one lucky validation split. Afterwards, use the untouched test set for the final report.

Functions and terms: Fold - One subset used for temporary validation; CV score - Average evaluation across held-out folds; Error analysis - Investigating cases the model predicts incorrectly

Common mistake: Do not cross-validate after repeatedly tuning on the final test set. Keep that test set reserved.